UNIT 6 Energy and Momentum of Rotating Systems

6.1 Rotational Kinetic Energy

6.2 Torque and Work

6.3 Angular Momentum and Angular Impulse

6.4 Conservation ofAngular Momentum

6.5 Rolling

6.6 Motion of Orbiting Satellites

6.1 Rotational Kinetic Energy

Kinetic energy

To make a body which is initially at rest rotate about a fixed axis, it is necessary to apply a torque to the body. The torque does work on the body and, as long as the applied torque exceeds the frictional torque, the work done increases the kinetic energy of the body and the faster the body rotates.

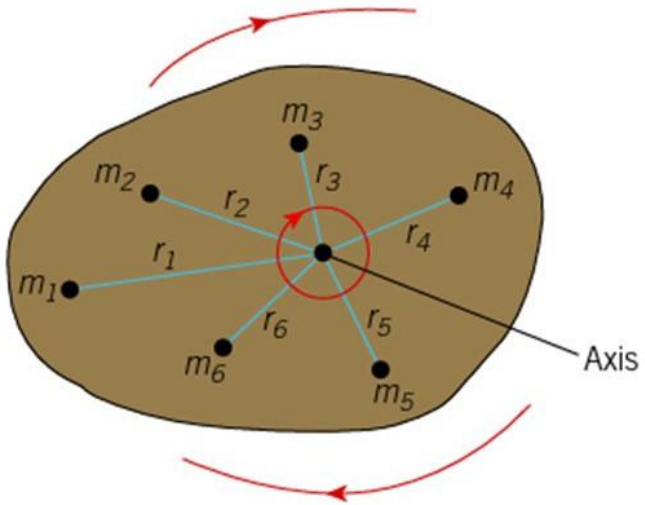
The kinetic energy, E_K, of a body rotating at angular speed ω is given by

$$E_K = \frac{1}{2} I \omega^2$$

where I is its moment of inertia about the axis of rotation.

To prove this equation, consider the body as a network of point masses m_1, m_2, m_3, m_4, etc. When the body rotates at angular speed ω, the speed of each point mass is given by $v = \omega r$ (see Figure 2 in Topic 25.2). So

speed of $m_1 = \omega r_1$, where r_1 is the distance of m_1 from the axis



▲ **Figure 2** *A rigid body considered as a network of point masses*

speed of $m_2 = \omega r_2$, where r_2 is the distance of m_2 from the axis, etc.

.....

speed of $m_N = \omega r_N$, where r_N is the distance of m_N from the axis.

Since the kinetic energy of each point mass is given by $\frac{1}{2}mv^2$, then

kinetic energy of $m_1 = \frac{1}{2}m_1 v_1^2 = \frac{1}{2}m_1 \omega^2 r_1^2$,

kinetic energy of $m_2 = \frac{1}{2}m_2 v_2^2 = \frac{1}{2}m_2 \omega^2 r_2^2$, etc.

.....

kinetic energy of $m_N = \frac{1}{2}m_N v_N^2 = \frac{1}{2}m_N \omega^2 r_N^2$.

Therefore the total kinetic energy $= \frac{1}{2}m_1\omega^2 r_1{}^2 + \frac{1}{2}m_2\omega^2 r_2{}^2$

$$+ \ldots + \frac{1}{2}m_N\omega^2 r_N{}^2$$

$$= \frac{1}{2}[(m_1 r_1{}^2) + (m_2 r_2{}^2) + \ldots\ldots\ldots + (m_N r_N{}^2)]\omega^2$$

$$= \frac{1}{2}I\omega^2 \quad \text{because } I = [(m_1 r_1{}^2) + (m_2 r_2{}^2)$$

$$+ \ldots\ldots\ldots + (m_N r_N{}^2)]$$

The equation $E_K = \frac{1}{2}I\omega^2$ enables us to calculate how much energy a rotating object stores due to its rotational motion. In addition, it shows that the kinetic energy of a rotating object is proportional to

- its moment of inertia about the axis of rotation
- the square of its angular speed.

6.2 Torque and Work

Work done

The **work done W by a constant torque T** when the body is turned through angle θ is given by

$$W = T\theta$$

This can be seen by considering the torque as due to a force F acting at a perpendicular distance d from the axis of rotation. The force acts through a distance $s = \theta d$ when it turns the body through angle θ. Therefore, the work done by the force $W = Fs = Fd\theta = T\theta$ as $T = Fd$.

Assuming there is no frictional torque, the applied torque $T = I\alpha$.

Therefore, the work done W by the applied torque is given by $W = T\theta = (I\alpha)\theta$.

Because the dynamics equation for angular motion $\omega^2 = \omega_0{}^2 + 2\alpha\theta$ gives $\alpha\theta = \frac{1}{2}\omega^2 - \frac{1}{2}\omega_0{}^2$, then $W = (I\alpha)\theta = I(\frac{1}{2}\omega^2 - \frac{1}{2}\omega_0{}^2) = \frac{1}{2}I\omega^2 - \frac{1}{2}I\omega_0{}^2 =$ the gain of kinetic energy.

Therefore, in the absence of friction, the work done by the torque is equal to the gain of rotational kinetic energy of the body.

Power

For a body rotating at constant angular speed ω, the power P delivered by a torque T acting on the body is given by

$$P = T\omega$$

We can see how this equation arises by considering a constant torque T acting on a body for a time t.

If the body turns through an angle θ in this time, the work done W by the torque is given by $W = T\theta$.

Since the power P delivered by the torque is the rate of work done by the torque, then $P = \frac{W}{t}$.

Therefore $P = \frac{W}{t} = \frac{T\theta}{t} = T\omega$, where the angular speed $\omega = \frac{\theta}{t}$.

If a rotating body is acted on by an applied torque T_1 which is equally opposed by a frictional torque T_F, the resultant torque is zero so its angular speed ω is constant.

- The power P delivered by the applied torque $T = \frac{\text{work done } W}{\text{time taken } t}$ $= \frac{T\theta}{t} = T\omega$.
- The work done per second by the frictional torque $T_F = \frac{T_F\theta}{t} = T_F\omega$.

In this situation, the rate of transfer of energy due to the applied force is equal to the rate of transfer of energy to the surroundings by the frictional force. So the rotating body does not gain any kinetic energy.

Worked example

A flywheel is rotating at an angular speed of $120\,\text{rad}\,\text{s}^{-1}$ on a fixed axle which is mounted on frictionless bearings. The moment of inertia of the flywheel and the axle about the axis of rotation is $0.068\,\text{kg}\,\text{m}^2$.

a Calculate the rotational kinetic energy of the flywheel when it rotates at $120\,\text{rad}\,\text{s}^{-1}$.

b When a braking torque of $1.4\,\text{N}\,\text{m}$ is applied to its rim, the flywheel is brought to rest.

Calculate the number of turns the flywheel makes as it decelerates to a standstill.

Solution

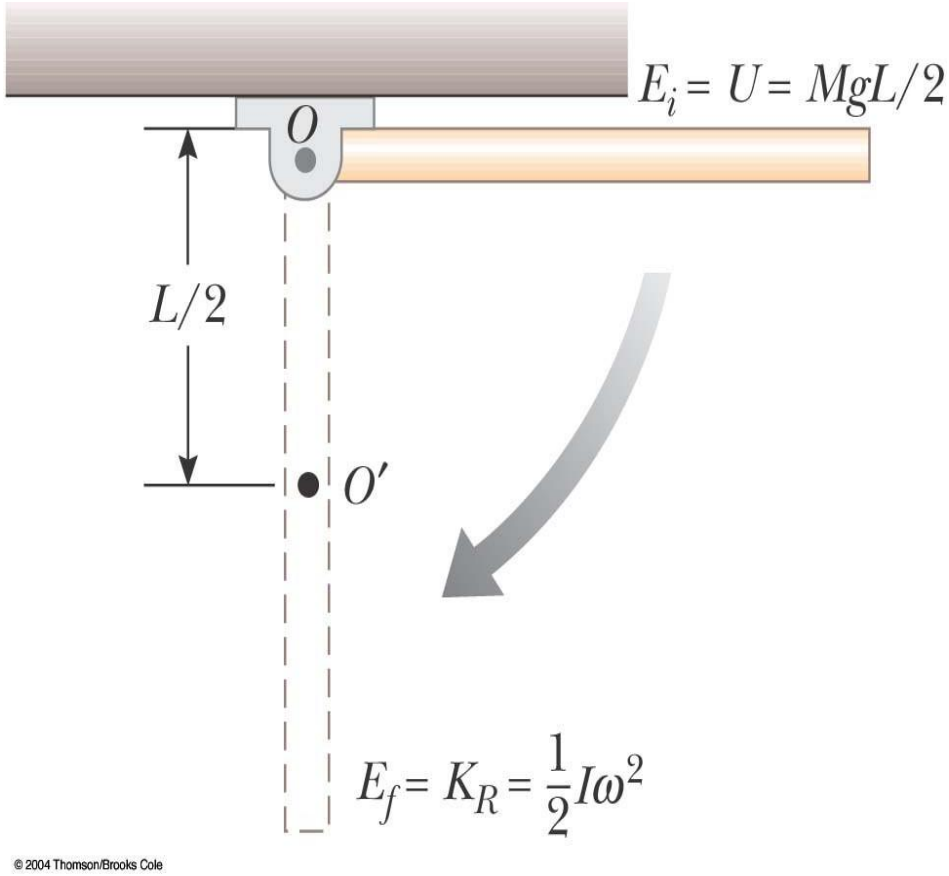
a $E_K = \frac{1}{2} I \omega^2 = 0.5 \times 0.068 \times 120^2 = 490\,\text{J}$

b The work done by the braking torque = loss of kinetic energy, ΔE_K

Therefore $T\theta = \Delta E_K$ so $1.4\theta = 490$

Therefore $\theta = \frac{490}{1.4} = 350\,\text{rad} = \frac{350}{2\pi} = 56$ turns

A uniform rod of length L and mass M is free to rotate on a frictionless pin passing through one end. The rod is released from rest in the horizontal position.



a) What is its angular speed when it reaches its lowest position ?

b) Determine the tangential speed of the center of mass and the tangential speed of the lowest point on the rod when it is in the vertical position.

6.3 Angular Momentum and Angular Impulse

Spin at work

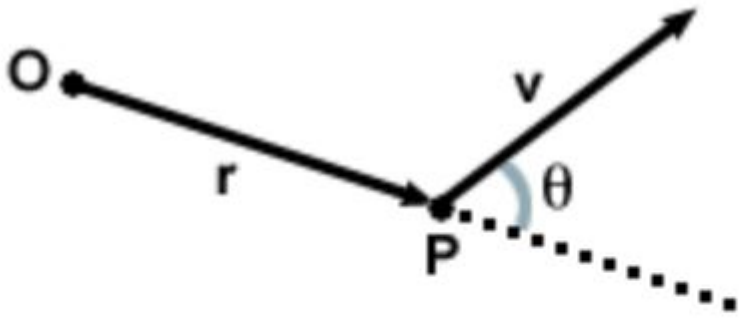
▲ **Figure 1** *A spinning ice skater*

© Patrick Giardino/Corbis

Tightly curling her body, a diver decreases her moment of inertia, increasing her angular velocity.

Angular Momentum for a Single Particle

Consider a single particle of mass **m** travelling with a velocity ***v*** and a radius ***r*** from an axis, as shown below.



A single particle moving with respect to an axis, O

The angular momentum of the single particle, then, is defined as:

$$l = rmv \sin\theta$$

Angular Momentum and Net Torque.

consider a rotating body with no resultant torque on it. Provided its moment of inertia stays the same, then its angular speed ω does not change. This can be seen by rewriting the equation $T = I\alpha$ where α is the angular acceleration. If the resultant torque T is zero, then the angular acceleration α is zero so the angular speed is constant.

In more general terms, the equation $T = I\alpha$ may be written as

$$T = \frac{\mathrm{d}}{\mathrm{d}t}(I\omega)$$

where $\frac{\mathrm{d}}{\mathrm{d}t}$ is the mathematical way of writing 'change per unit time',

Angular momentum

The quantity $I\omega$ is the **angular momentum** of the rotating body. So the above equation tells us that resultant torque T is equal to the rate of change of angular momentum of the rotating object.

Angular momentum of a rotating object = $I\omega = L$

where I is the moment of inertia of the body about the axis of rotation and ω is its angular speed. The unit of angular momentum is $\mathrm{kg\,m^2\,rad\,s^{-1}}$ or $\mathrm{N\,m\,s}$

When the resultant torque is zero, then $\frac{\mathrm{d}}{\mathrm{d}t}(I\omega) = 0$ which means its angular momentum $I\omega$ is constant.

Angular momentum

In the ice skater example, the moment of inertia of the ice skater suddenly decreases when she pulls the arms in. Since the angular momentum is constant, the sudden decrease in the moment of inertia causes the angular speed to increase. In specific terms, if the moment of inertia changes from I_1 to I_2 the angular speed changes from ω_1 to ω_2 such that

$$I_1\omega_1 = I_2\omega_2$$

Angular momentum

Notes:

For a rotating object whose moment of inertia I does **not** change:

1 $T = \frac{d}{dt}(I\omega) = I\frac{d\omega}{dt} = I\alpha$ because the angular acceleration $\alpha = \frac{d\omega}{dt}$.

2 If the object undergoes uniform angular acceleration from rest to reach angular speed ω in time t, the resultant torque acting on it $T = \frac{I\omega}{t}$. Therefore its angular momentum $I\omega = Tt$. This equation shows that the unit of angular momentum can also be given as the unit of torque × the unit of time (i.e., N m s).

Angular impulse

When the resultant torque T on a rotating object acts for a short time interval Δt, the angular velocity of the object changes rapidly. This can happen for example when a bat or a racquet hits a moving ball, changing the angular momentum or spin of the ball as well as its direction of motion.

The **angular impulse** of a short-duration torque is defined as $\boldsymbol{T\Delta t}$. If $\Delta\omega$ is the change of angular momentum due to this torque T, then

$$\textbf{angular impulse } T\Delta t = \textbf{change of angular momentum } I\Delta\omega$$

where I is the moment of inertia of the object about the axis of rotation.

6.4 Conservation of Angular Momentum

The total angular momentum of a system is constant in both magnitude and direction if the resultant external torque acting on the system is zero, that is, if the system is isolated.

$$\boxed{L_{before} = L_{after}}$$

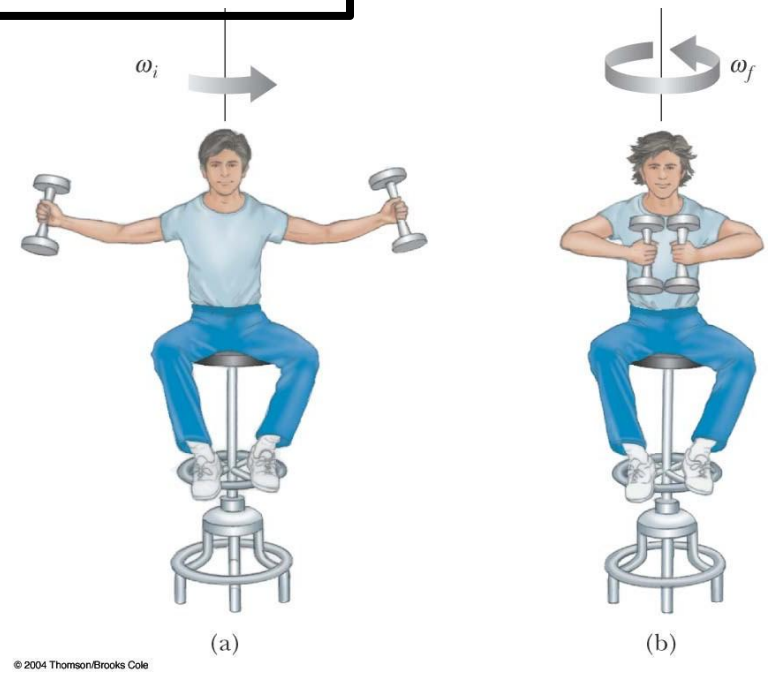
$L_{before} = L_{after}$

Since $L = Iw$,

$$I_i w_i = I_f w_f = \text{constant}$$

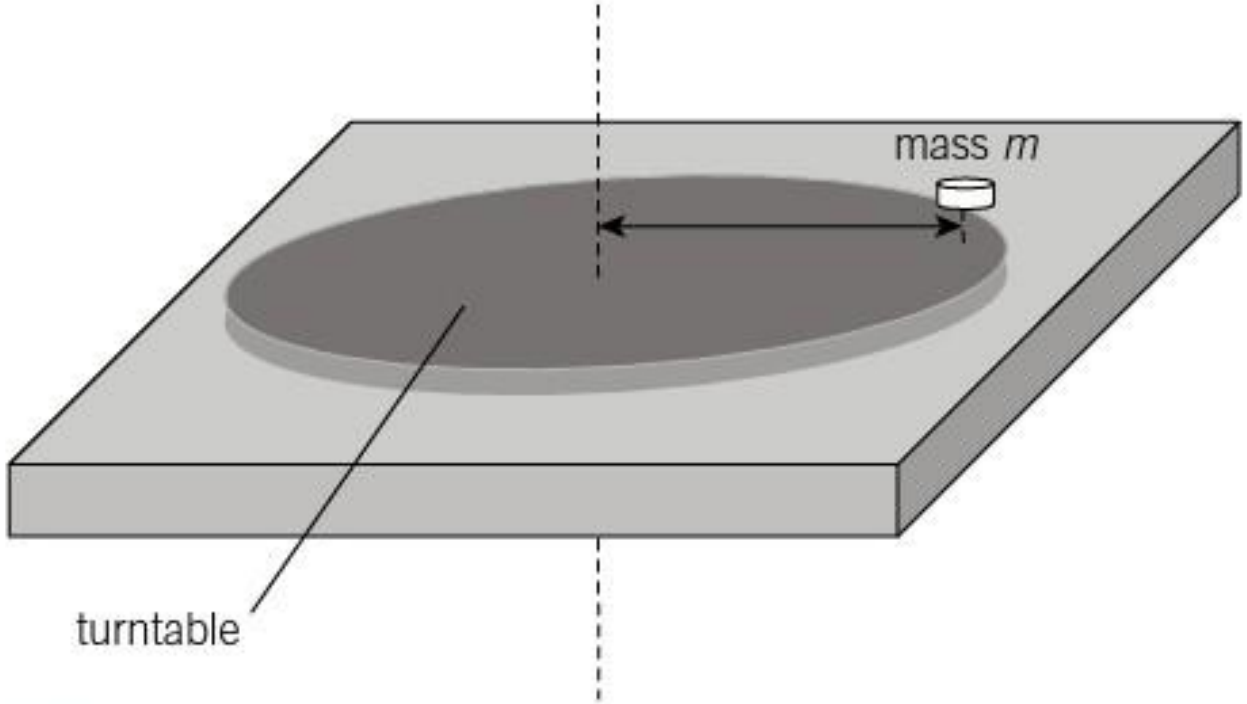
When arms are moved inward, I decreases.

Since Iw remains constant, as I decreases, w must increase.



Conservation of angular momentum

2 **If a small object dropped onto a freely rotating turntable,** as shown in Figure 4, so that it sticks to the turntable, the object gains angular momentum and the turntable loses angular momentum.



▲ **Figure 4** *Measuring the moment of inertia of a freely rotating turntable*

Conservation of angular momentum

- The total angular momentum of the turntable before the object is dropped onto it = $I_1\omega_1$, where I_1 is the moment of inertia of the turntable and ω_1 is its initial angular speed.
- The total angular momentum after the object has been dropped onto the turntable = $(I_1 + I_2)\omega_2$, where ω_2 is the final angular speed and $(I_1 + I_2)$ is the total moment of inertia about the axis of rotation of the turntable.
- If the mass of the object is m and its perpendicular distance from the axis of rotation of the turntable is r, then the moment of inertia I_2 of the object about the axis of rotation = mr^2.

Conservation of angular momentum

According to the conservation of momentum, $(I_1 + mr^2)\omega_2 = I_1\omega_1$.

Rearranging this equation gives $I_1 = \dfrac{mr^2 \times \omega_2}{\omega_1 - \omega_2}$.

Therefore I_1 can be found by measuring m, r, ω_1, and ω_2.

Note:

The angular momentum of a point mass is defined as its momentum × its distance from the axis of rotation. For a point mass m rotating at angular speed ω at distance r from the axis, its momentum is $m\omega r$ (because its speed $v = \omega r$) so its angular momentum is $m\omega r^2$.

For a network of point masses $m_1, m_2, \ldots, m_N$ which make up a rigid body, the total angular momentum

$$= (m_1 r_1^2 \omega) + (m_2 r_2^2 \omega) + \ldots + (m_N r_N^2 \omega)$$

$$= [(m_1 r_1^2) + (m_2 r_2^2) + \ldots + (m_N r_N^2)]\omega = I\omega$$

Example 11.2 The Torque Vector

A force of $\vec{\mathbf{F}} = (2.00\,\hat{\mathbf{i}} + 3.00\,\hat{\mathbf{j}})$ N is applied to an object that is pivoted about a fixed axis aligned along the z coordinate axis. The force is applied at a point located at $\vec{\mathbf{r}} = (4.00\,\hat{\mathbf{i}} + 5.00\,\hat{\mathbf{j}})$ m. Find the torque $\vec{\boldsymbol{\tau}}$ applied to the object.

Example 11.3 Angular Momentum of a Particle in Circular Motion

A particle moves in the xy plane in a circular path of radius r as shown in Figure 11.5. Find the magnitude and direction of its angular momentum relative to an axis through O when its velocity is $\vec{\mathbf{v}}$.

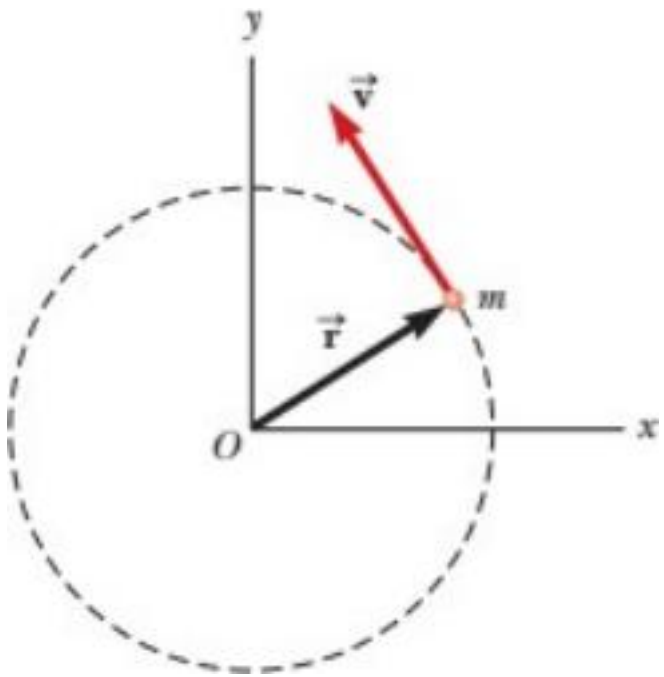
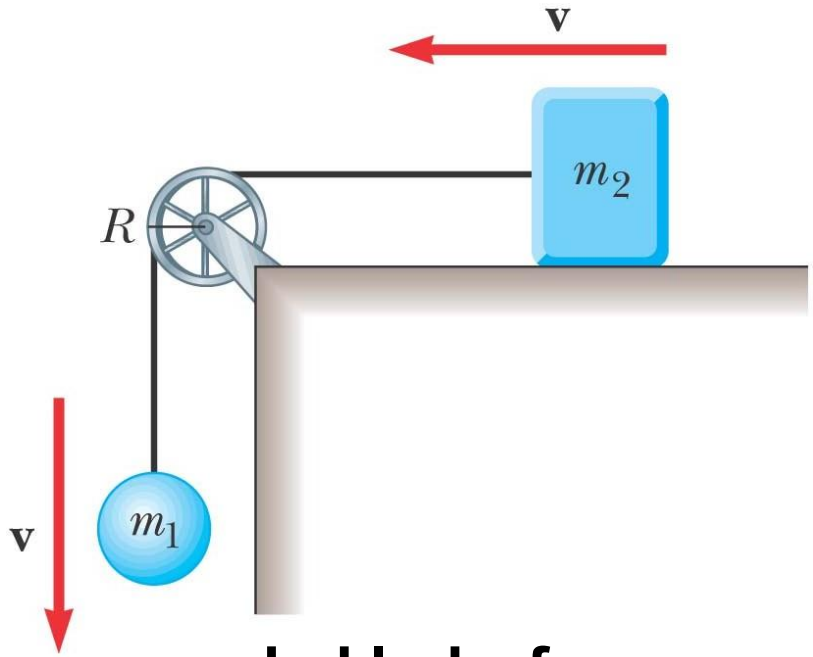


Figure 11.5 (Example 11.3) A particle moving in a circle of radius r has an angular momentum about an axis through O that has magnitude mvr. The vector $\vec{\mathbf{L}} = \vec{\mathbf{r}} \times \vec{\mathbf{p}}$ points *out* of the page.

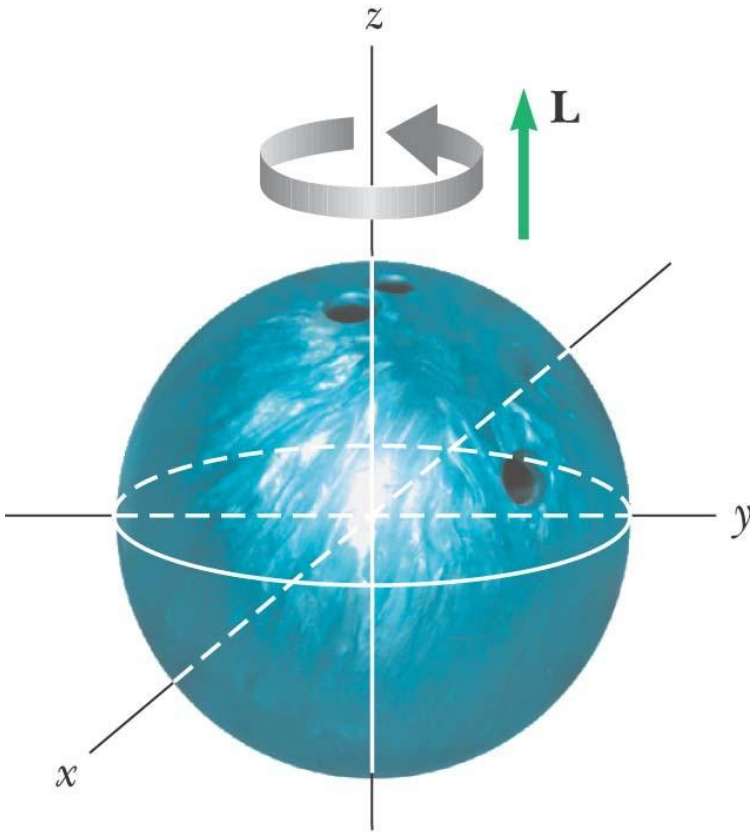
Example 11.4



© 2004 Thomson/Brooks Cole

A sphere of mass m_1 and a block of mass m_2 are connected by a light cord that passes over a pulley, as shown above. The radius of the pulley is *R*, and the mass of the rim is *M*. The spokes of the pulley have negligible mass. The block slides on a frictionless, horizontal surface. Find an expression for the linear acceleration of the two objects, using the concepts of angular momentum and torque.

Example 11.5

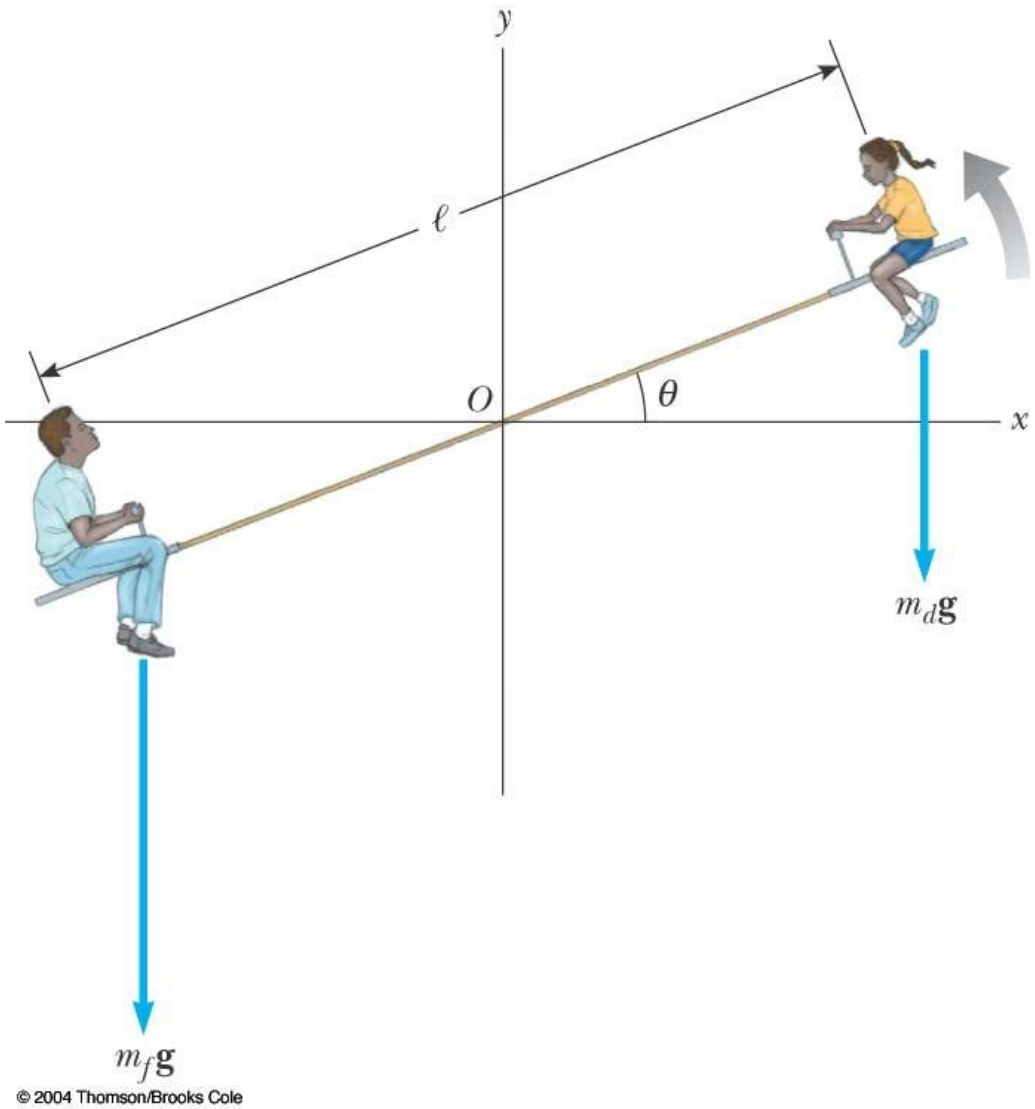


© 2004 Thomson/Brooks Cole

Estimate the magnitude of the angular momentum of a bowling ball spinning at 10 rev/s.

Example 11.6

A father of mass mf and daughter of mass md sit on opposite ends of a seesaw at equal distances from the pivot at the center. The seesaw is modeled as a rigid rod of mass M and length l and is pivoted without friction. At a given moment, the combination rotates in a vertical plane with an angular speed w.



a) Find an expression for the magnitude of the system's angular momentum.

b) Find an expression for the magnitude of the angular acceleration of the system when the seesaw makes an angle q with the horizontal.

Example 11.8 The Merry-Go-Round AM

A horizontal platform in the shape of a circular disk rotates freely in a horizontal plane about a frictionless, vertical axle (Fig. 11.11). The platform has a mass $M = 100$ kg and a radius $R = 2.0$ m. A student whose mass is $m = 60$ kg walks slowly from the rim of the disk toward its center. If the angular speed of the system is 2.0 rad/s when the student is at the rim, what is the angular speed when she reaches a point $r = 0.50$ m from the center?

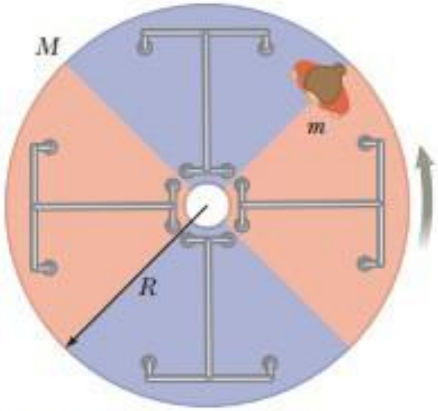


Figure 11.11 (Example 11.8) As the student walks toward the center of the rotating platform, the angular speed of the system increases because the angular momentum of the system remains constant.

Example 11.9 Disk and Stick Collision AM

A 2.0-kg disk traveling at 3.0 m/s strikes a 1.0-kg stick of length 4.0 m that is lying flat on nearly frictionless ice as shown in the overhead view of Figure 11.12a. The disk strikes at the endpoint of the stick, at a distance $r = 2.0$ m from the stick's center. Assume the collision is elastic and the disk does not deviate from its original line of motion. Find the translational speed of the disk, the translational speed of the stick, and the angular speed of the stick after the collision. The moment of inertia of the stick about its center of mass is $1.33\ \text{kg} \cdot \text{m}^2$.

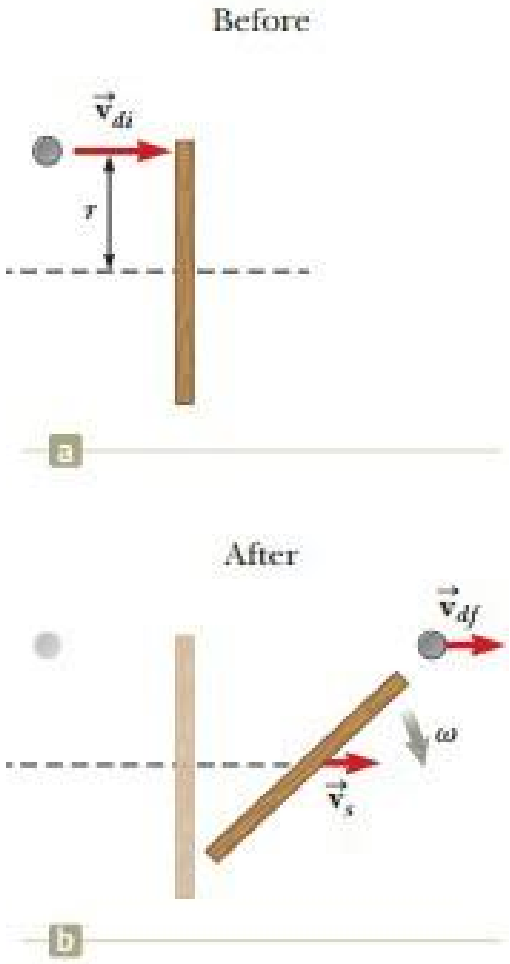
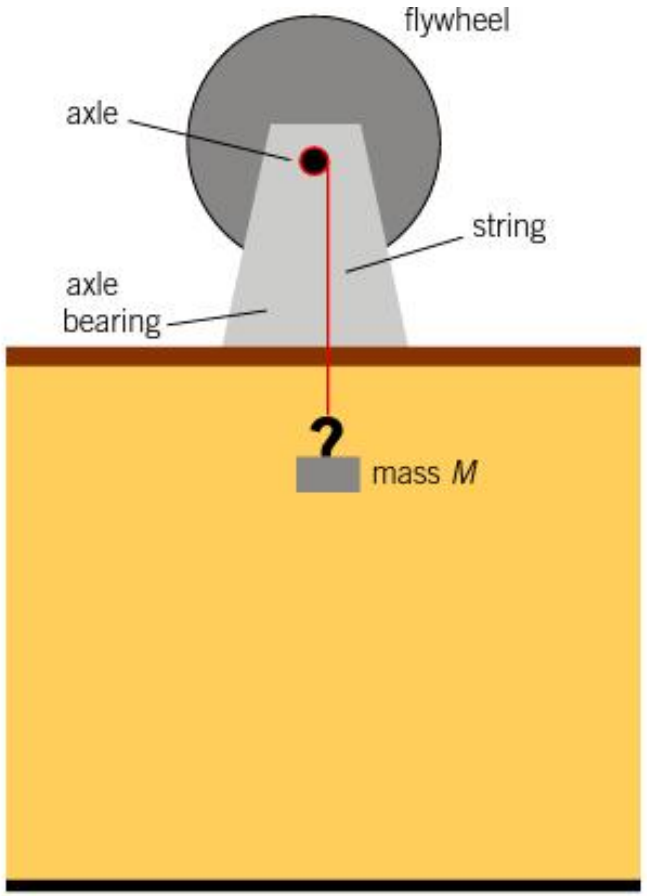


Figure 11.12 (Example 11.9) Overhead view of a disk striking a stick in an elastic collision. (a) Before the collision, the disk moves toward the stick. (b) The collision causes the stick to rotate and move to the right.

6.5 Rolling



▲ Figure 1

Experiment to measure the moment of inertia of a flywheel

An object of known mass M hanging from a string is used to accelerate the flywheel from rest, as shown in Figure 1.

The following measurements need to be made:

1. the distance fallen, h, by the object from release to when the string unwraps itself from the axle of the flywheel
2. the diameter d of the axle
3. the time taken t for the string to unwrap.

Experiment to measure the moment of inertia of a flywheel

The measurements and the known mass M are used in the following calculations:

- the number of turns made by the flywheel as the string unwraps, $N = \frac{h}{\pi d}$
- the maximum angular speed of the flywheel, $\omega_{max} = 2 \times$ the average angular speed $= \frac{2 \times (2\pi N)}{t}$
- the speed of the object at the instant the string unwraps, $v = \frac{\omega d}{2}$
- the kinetic energy gained by the flywheel, $\Delta E_{KF} = \frac{1}{2} I\omega^2$, to be calculated in terms of I

Experiment to measure the moment of inertia of a flywheel

- the kinetic energy gained by the object of mass M, $\Delta E_{KO} = \frac{1}{2}Mv^2$ where $v = \frac{\omega d}{2}$
- the potential energy lost by the object of mass M, $\Delta E_P = Mgh$.

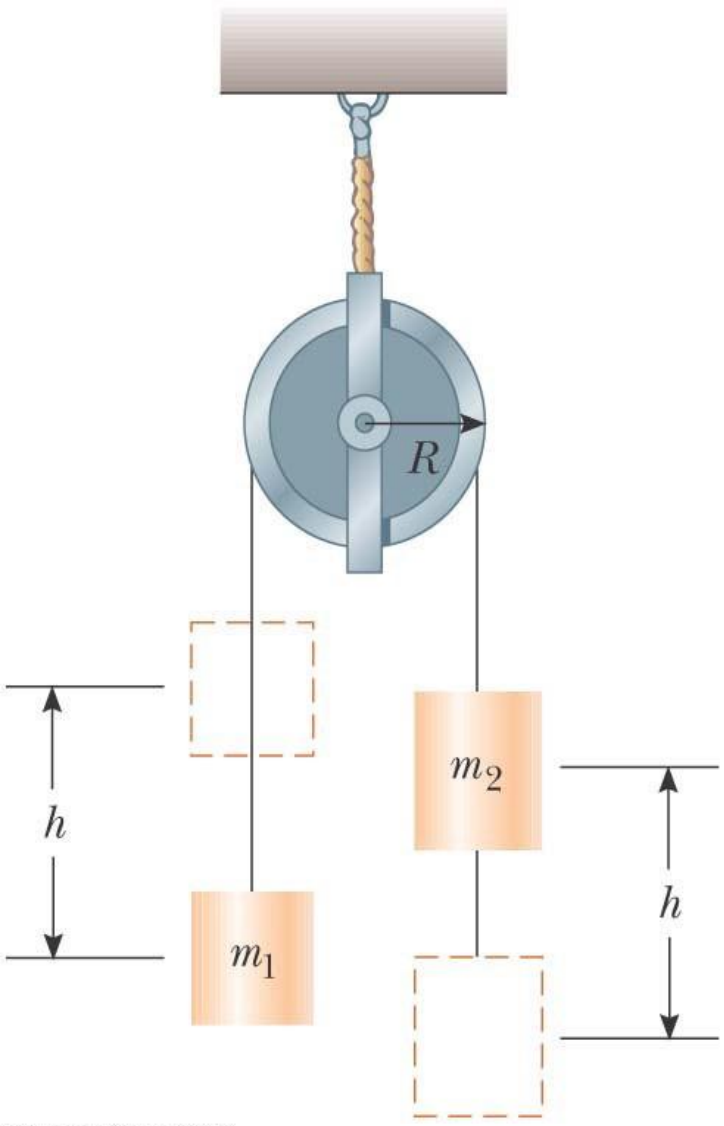
The moment of inertia of the flywheel can be calculated from the equation below, assuming friction on the flywheel is negligible.

The total gain of kinetic energy, $\Delta E_{KF} + \Delta E_{KO}$ = the loss of potential energy, ΔE_P

$$\frac{1}{2}I\omega^2 + \frac{1}{2}Mv^2 = Mgh$$

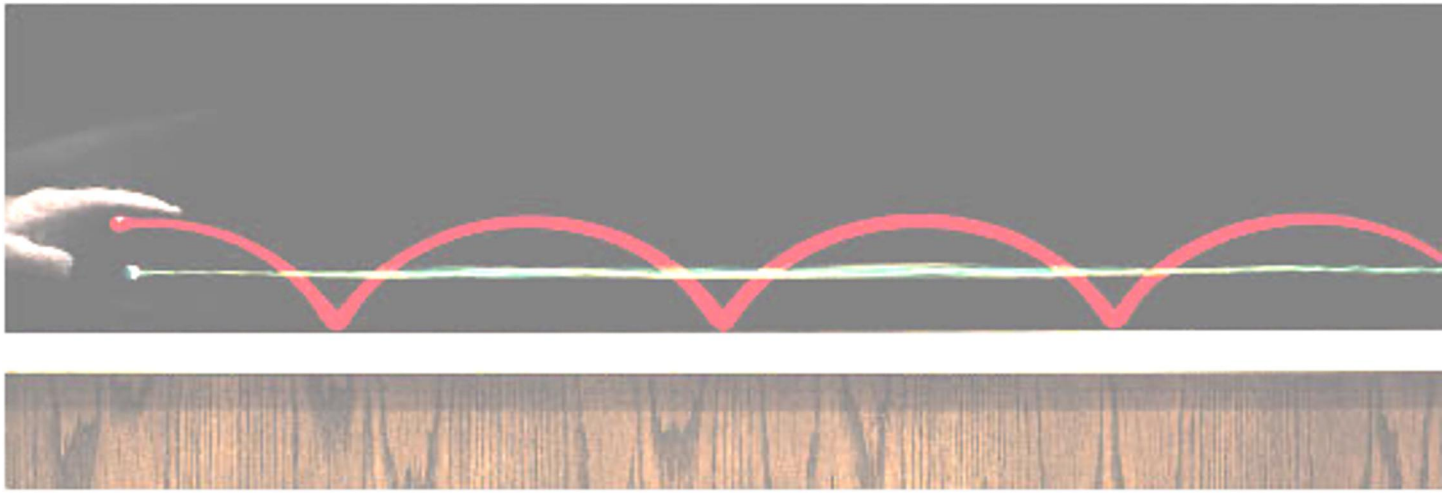
Example 10.12

Consider two cylinders having different masses m_1 and m_2, connected by a string passing over a pulley. The pulley has a radius *R* and moment of inertia *I* about its axis of rotation. The string does not slip on the pulley, and the system is released from rest. Find the linear (translational) speeds of the cylinders after cylinder 2 descends through a distance *h*, and the angular speed of the pulley at this time.



© 2004 Thomson/Brooks Cole

Rolling Motion of a Rigid Object



Cylinder rolling on a straight path.

Center moves in a straight line (green line).

A point on the rim moves in a path called a cycloid (red curve).

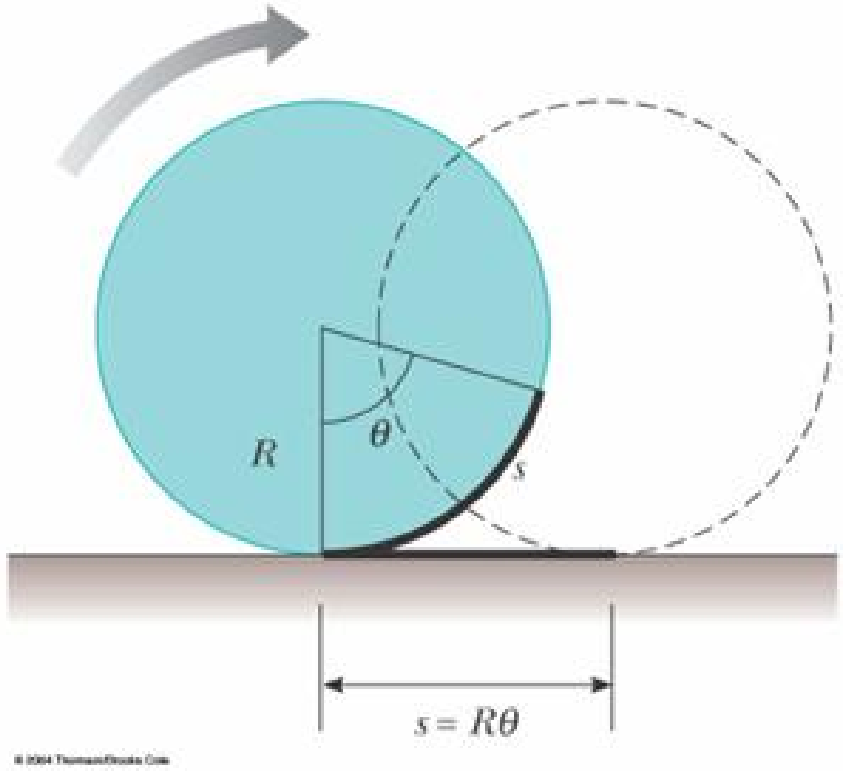
Speed of CM of Cylinder Rolling without Slipping

$$v_{CM} = \frac{ds}{dt}$$

Since $s = R\theta$,

$$v_{CM} = R\,\frac{d\theta}{dt}$$

$$\boxed{v_{CM} = R\omega}$$



for pure rolling motion only

Acceleration of CM of Cylinder Rolling without Slipping

$$a_{CM} = \frac{dv_{CM}}{dt}$$

$$a_{CM} = R\,\frac{d\omega}{dt}$$

$$\boxed{a_{CM} = R\alpha}$$

Total Kinetic Energy of a Rolling Cylinder

$$\boxed{KE_{rot} = \tfrac{1}{2} I_P \omega^2}$$

Since $I_P = I_{CM} + MR^2$, (parallel-axis theorem)

$$KE_{rot} = \tfrac{1}{2} (I_{CM} + MR^2)\omega^2$$

OR

$$KE_{rot} = \tfrac{1}{2} I_{CM}\omega^2 + \tfrac{1}{2} MR^2\omega^2$$

Since $v_{CM} = R\omega$,

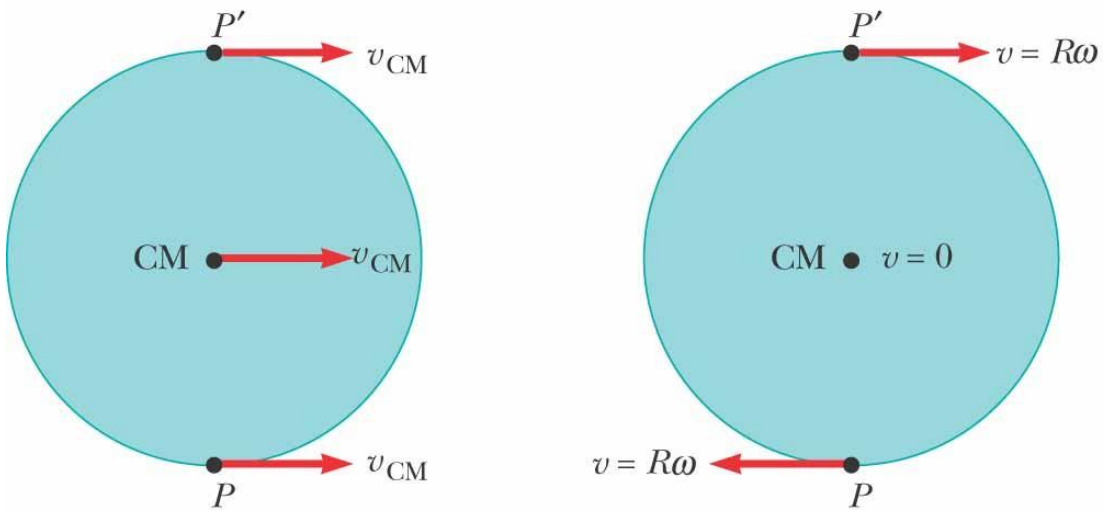
$$\boxed{KE_{rot} = \tfrac{1}{2} I_{CM}\omega^2 + \tfrac{1}{2} Mv_{CM}^2}$$

$$KE_{total} = \tfrac{1}{2} I_{CM}\omega^2 + \tfrac{1}{2} M v_{CM}^2$$

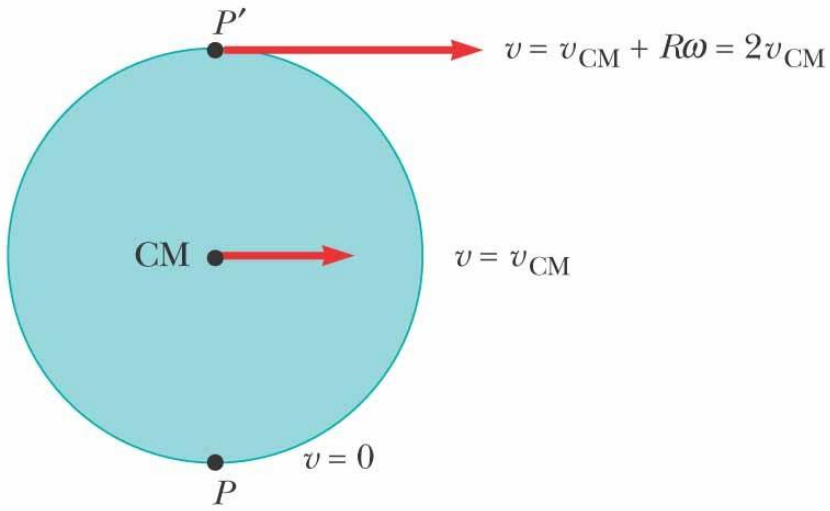
rotational KE

translational KE

The total kinetic energy of a rolling object is the sum of the *rotational* kinetic energy about the center of mass and the *translational* kinetic energy of the center of mass.



(a) Pure translation (b) Pure rotation



(c) Combination of translation and rotation

© 2004 Thomson/Brooks Cole

Example 10.13 Sphere Rolling Down an Incline AM

For the solid sphere shown in Figure 10.26, calculate the translational speed of the center of mass at the bottom of the incline and the magnitude of the translational acceleration of the center of mass.

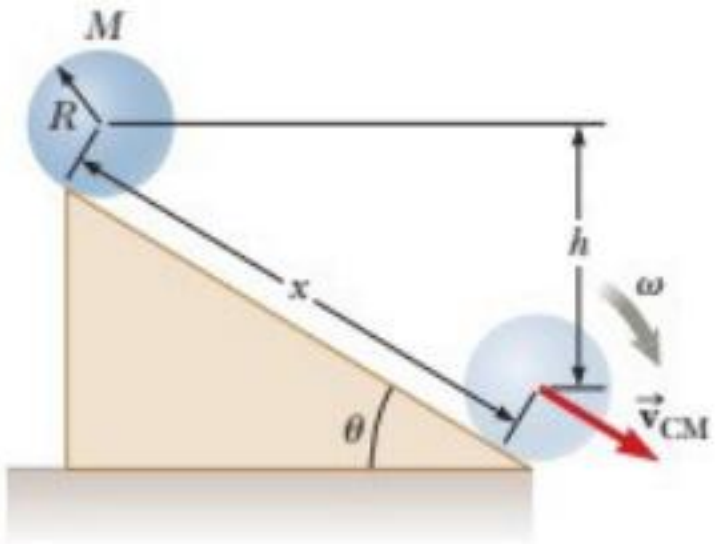


Figure 10.26 A sphere rolling down an incline. Mechanical energy of the sphere–Earth system is conserved if no slipping occurs.

SOLUTION

Apply conservation of energy with $PE = PE_g$, the potential energy associated with gravity:

$$(KE_t + KE_r + PE_g)_i = (KE_t + KE_r + PE_g)_f$$

Substitute the appropriate general expressions, noting that $(KE_t)_i = (KE_r)_i = 0$ and $(PE_g)_f = 0$ (obtain the moment of inertia of a ball from Table 8.1):

$$0 + 0 + Mgh = \tfrac{1}{2}Mv^2 + \tfrac{1}{2}(\tfrac{2}{5}MR^2)\omega^2 + 0$$

The ball rolls without slipping, so $R\omega = v$, the "no-slip condition," can be applied:

$$Mgh = \tfrac{1}{2}Mv^2 + \tfrac{1}{5}Mv^2 = \tfrac{7}{10}Mv^2$$

$$v = \sqrt{\frac{10gh}{7}}$$

Example 10.14 Pulling on a Spool[3] AM

A cylindrically symmetric spool of mass m and radius R sits at rest on a horizontal table with friction (Fig. 10.27). With your hand on a light string wrapped around the axle of radius r, you pull on the spool with a constant horizontal force of magnitude T to the right. As a result, the spool rolls without slipping a distance L along the table with no rolling friction.

(A) Find the final translational speed of the center of mass of the spool.

(B) Find the value of the friction force f.

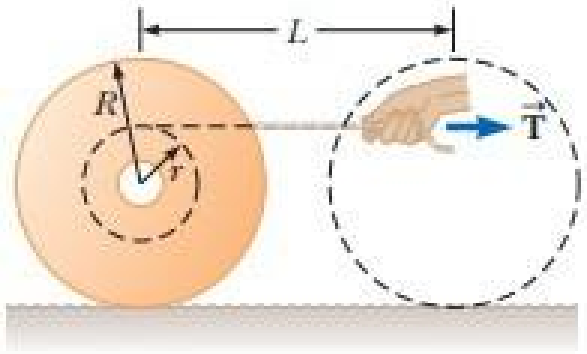


Figure 10.27 (Example 10.14) A spool rests on a horizontal table. A string is wrapped around the axle and is pulled to the right by a hand.

6.6 Motion of Orbiting Satellites

- **A geostationary orbit is one in which a satellite will orbit with exactly the same period of rotation as the Earth (24 hours).**
- **Geostationary satellites have same period of rotation as the earth and move in the same direction as the earth so that they remain above a fixed point on the Earth's equator.**
- **They are used for telecommunications (transmitting telephone messages around the world) and for satellite television transmission.**

The radius of orbit of a geostationary satellite can be calculated using the equation $\frac{r^3}{T^2} = \frac{GM}{4\pi^2}$.

So, $T = 24\,\text{h} = 24 \times 3600\,\text{s} = 86\,400\,\text{s}$

$$r^3 = \frac{GMT^2}{4\pi^2} = \frac{6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times (86\,400)^2}{4\pi^2} = 7.6 \times 10^{22}\,\text{m}^3$$

$r = 4.2 \times 10^7\,\text{m} = 42\,000\,\text{km}$.

The radius of the Earth is 6400 km. Therefore, the height of a geostationary satellite above the Earth is 36 000 km (= 42 000 − 6400 km).

Note:

That a geosynchronous orbit is a 24 hour orbit inclined to the equator.

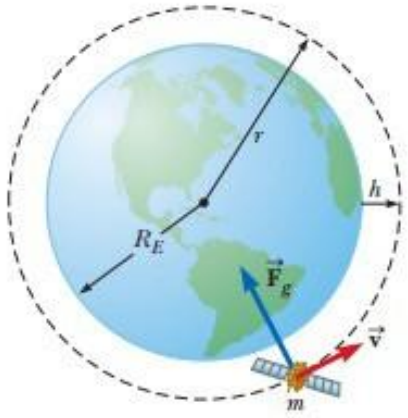


Figure 13.9 (Example 13.5) A satellite of mass m moving around the Earth in a circular orbit of radius r with constant speed v. The only force acting on the satellite is the gravitational force $\vec{F}_g$. (Not drawn to scale.)

Example 13.5 A Geosynchronous Satellite AM

Consider a satellite of mass m moving in a circular orbit around the Earth at a constant speed v and at an altitude h above the Earth's surface as illustrated in Figure 13.9.

(A) Determine the speed of satellite in terms of G, h, R_E (the radius of the Earth), and M_E (the mass of the Earth).

(B) If the satellite is to be *geosynchronous* (that is, appearing to remain over a fixed position on the Earth), how fast is it moving through space?

Gravitational potential

- **Imagine you are in a space rocket, about to blast off from the surface of a planet. The planet's gravitational field extends far into space although it becomes weaker with increased distance from the planet.**
- **To escape from the planet's pull due to gravity, the rocket must do work against force of gravity on it due to the planet.**
- **The planet's gravitational field is like a trap which the rocket must climb out from to escape.**
- **As the rocket rises, it gravitational potential energy (g.p.e) increases.**
- **If it falls back it gravitational potential energy decreases.**

Gravitational potential

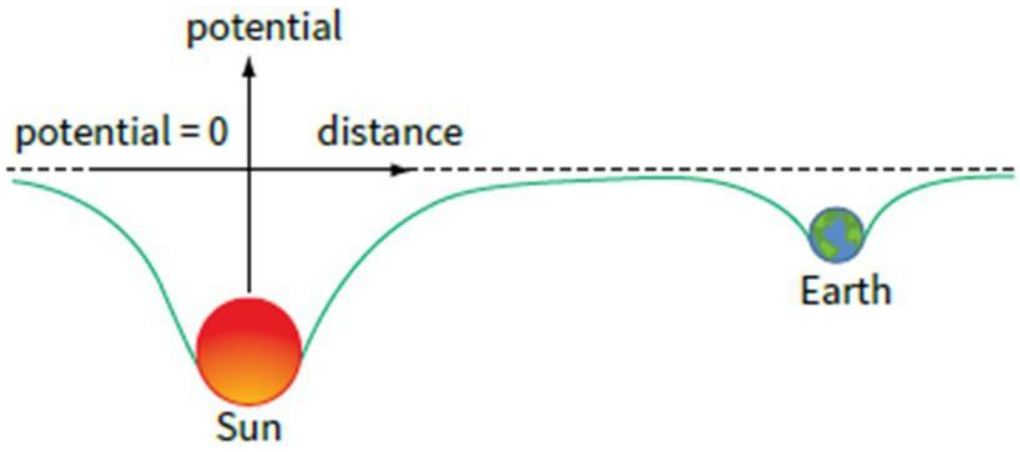


Figure 18.8 The gravitational potential is zero at infinity (far from any mass), and decreases as a mass is approached.

- **If the rocket carried a g.p.e meter which was set at zero when it was far from away from the planet, the meter reading would be negative when the rocket is on the surface of the planet and would increase towards zero as the rocket moves away from the planet.**

Gravitational potential energy

- **Gravitational potential energy is the energy of an object due to its position in a gravitational field.**
- **The position of zero gravitational potential energy is at infinity – in other words, the object would be so far away that the gravitational force on it is negligible.**
- **Your rocket climbing out of the planet's field needs to increase its g.p.e to zero to escape completely.**
- **At the surface, its g.p.e is negative, so needs to do work to escape from the field completely.**

Gravitational Potential Energy

$$\boxed{U_g = mgh}$$

only valid when the mass is near the Earth's surface, where the gravitational force is constant

What is the *general form* of the gravitational potential energy function ?

Gravitational potential

- The **gravitational potential** at a point is the work done per unit mass in bringing a mass from infinity to the point.
- The symbol used for potential is V, thus

$$\boldsymbol{U} = \frac{\boldsymbol{W}}{\boldsymbol{m}}$$

The unit for $\boldsymbol{U}$ is Jkg^{-1}.

Gravitational potential

- Because the work done ΔW to move a rocket from U_1 to U_2 is equal to the change of its gravitaional potential energy, then $\Delta W = \mathrm{m}\Delta U$.
- In general, if a small object of mass m is moved from a gravitational potential U_1 to U_2, its change in gravitaional potential energy (g.p.e) is given as:

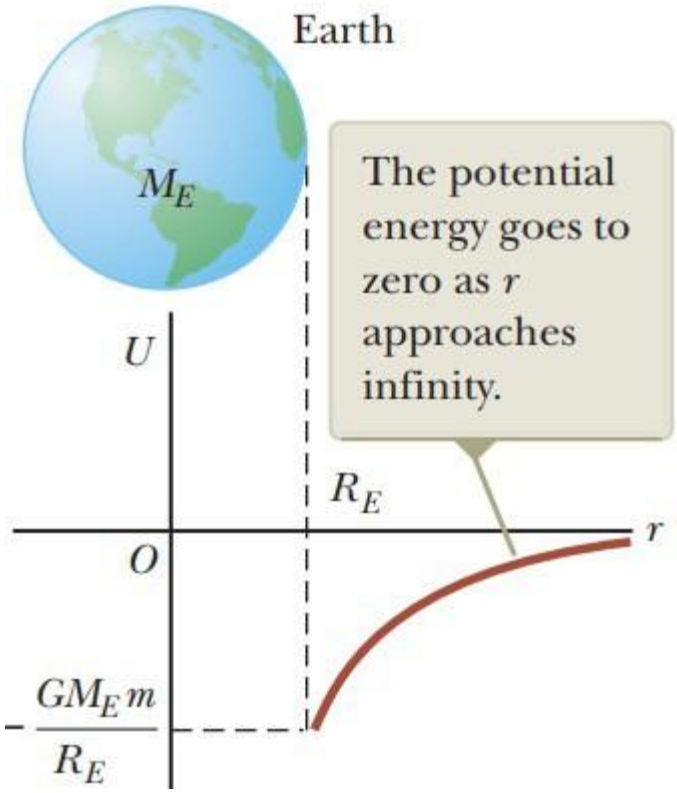
$$\Delta \boldsymbol{U_g} = \boldsymbol{m}\,(\boldsymbol{U_2} - \boldsymbol{U_1}) = \boldsymbol{m}\Delta \boldsymbol{V}$$

The variation of g with distance from the centre of a spherical planet (or star)

- Consider, the gravitational field beyond the surface of a spherical planet of mass M and radius R.
- As seen previously, for any position at or beyond the surface, the magnitude of the gravitational field strength is given by $\boldsymbol{g} = \frac{\boldsymbol{GM}}{\boldsymbol{r^2}}$, where r is the distance from the position to the centre of the sphere.
- The gravitational field strength at the surface of a sphere of radius R and mass M is given by $\boldsymbol{g_s} = \frac{\boldsymbol{GM}}{\boldsymbol{R^2}}$ because the distance from the centre, r, is equal to R at the surface.

The variation of g with distance from the centre of a spherical planet (or star)

- Because the gravitational field strength $g_s = \frac{GM}{R^2}$, then $GM = g_s R^2$.
- Therefore, $g = \frac{g_s R^2}{r^2}$
- The equation shows how g changes with increase of distance r from the centre of the planet.
  - At distance $r = 2R$, $g = \frac{g_s R^2}{(2R)^2} = \frac{g_s}{4}$
  - At distance $r = 3R$, $g = \frac{g_s R^2}{(3R)^2} = \frac{g_s}{9}$
  - At distance $r = 4R$, $g = \frac{g_s R^2}{(4R)^2} = \frac{g_s}{16}$



$$U = -\frac{Gm_1 m_2}{r}$$

In General (for any two particles)

$$U = -\frac{Gm_1m_2}{r}$$

Gravitational potential energy varies as 1/r, whereas gravitational force varies as $1/r^2$.

U becomes less negative as r increases.

U becomes zero when r is infinite.

Example 13.6 **The Change in Potential Energy**

A particle of mass m is displaced through a small vertical distance Δy near the Earth's surface. Show that in this situation the general expression for the change in gravitational potential energy given by Equation 13.13 reduces to the familiar relationship $\Delta U = mg\Delta y$.

A particle of mass m is displaced through a small vertical distance Dy near the Earth's surface. Show that in this situation the general expression for the change in gravitational potential energy given by

$$U_f - U_i = -GM_E m \left(\frac{1}{r_f} - \frac{1}{r_i} \right)$$

reduces to the familiar relationship $\Delta U = mg\Delta y$.

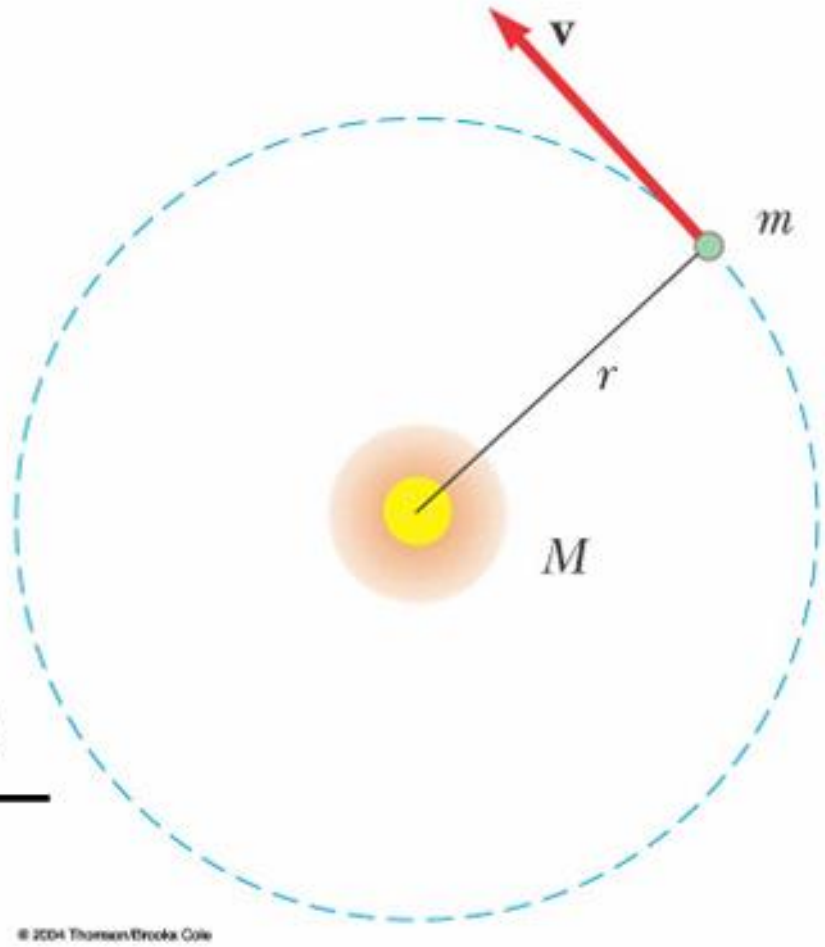
Energy in Planetary and Satellite Motion

$$E = KE + U$$

$$E = \frac{1}{2} mv^2 - \frac{GMm}{r}$$

Newton's Second Law applied to mass *m* :

$$\frac{GMm}{r^2} = ma = \frac{mv^2}{r}$$



© 2004 Thomson/Brooks Cole

Multiplying both sides by r and dividing by 2,

$$\frac{1}{2} mv^2 = \frac{GMm}{2r}$$

Substituting into $E = \frac{1}{2} mv^2 - \frac{GMm}{r}$,

$$E = \frac{GMm}{2r} - \frac{GMm}{r}$$

$$\boxed{E = -\frac{GMm}{2r}}$$

(total energy for circular orbits)

$$E = -\frac{GMm}{2r}$$

total energy is negative

KE is positive and equal to half the absolute value of U.

For elliptical orbits, we replace r with semimajor axis a,

$$\boxed{E = -\frac{GMm}{2a}}$$

(total energy for elliptical orbits)

Energy in planetary motion

The space shuttle releases a 470 kg communications satellite while in an orbit 280 km above the surface of the Earth. A rocket engine on the satellite boosts it into a geosynchronous orbit, which is an orbit in which the satellite stays directly over a single location on the Earth.

How much energy does the engine have to provide ?

Example 13.7 Changing the Orbit of a Satellite

A space transportation vehicle releases a 470-kg communications satellite while in an orbit 280 km above the surface of the Earth. A rocket engine on the satellite boosts it into a geosynchronous orbit. How much energy does the engine have to provide?

Escape velocity

- **Escape velocity** - Minimum value of the initial speed needed to allow the object to move infinitely far away from the Earth.
- At the surface of a planet of radius R, the potential

$U = -\frac{GM}{R}$.

- At infinity, the potential is zero. Therefore, to move an object of mass m from the surface to infinity, the work that must be done is:

$$\Delta W = m\Delta U = \frac{GMm}{R}$$

Escape velocity

- If the object is projected at speed v, then for it to be able to escape from the planet, its initial kinetic energy $\frac{1}{2}mv^2 \geq \Delta W$.

So, $\frac{1}{2}mv^2 \geq \Delta W$, which gives $\frac{1}{2}mv^2 \geq \frac{GMm}{R}$

Therefore, $v^2 \geq \frac{2GM}{R}$

So, the escape velocity, $v_{esc} = \sqrt{\frac{2GM}{R}}$

Because $g = \frac{GM}{R^2}$, substituting gR^2 for GM therefore gives

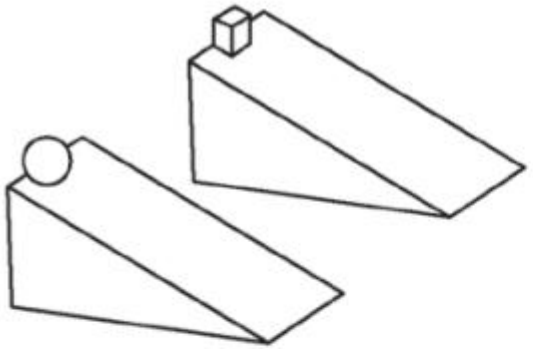
$$\boldsymbol{v}_{esc} = \sqrt{2gR}$$

Example 13.8 Escape Speed of a Rocket

Calculate the escape speed from the Earth for a 5 000-kg spacecraft and determine the kinetic energy it must have at the Earth's surface to move infinitely far away from the Earth.

Calculate the escape speed from the Earth for a 5000 kg spacecraft and determine the kinetic energy it must have at the Earth's surface in order to move infinitely far away from the Earth.

Example Question of AP Classroom



Two objects are released from rest at the top of ramps with the same dimensions, as shown in the figure above. The sphere rolls down one ramp without slipping. The small block slides down the other ramp without friction. Which object reaches the bottom of its ramp first, and why?

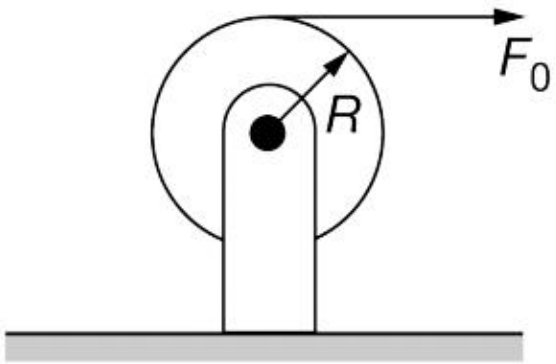
(A) The sphere, because it gains rotational kinetic energy, but the block does not

(B) The sphere, because it gains mechanical energy due to the torque exerted on it, but the block does not

(C) The block, because it does not lose mechanical energy due to friction, but the sphere does

(D) The block, because it does not gain rotational kinetic energy, but the sphere does

Example Question of AP Classroom



A uniform cylinder of radius R, mass M, and rotational inertia I_0 is initially at rest. The cylinder is mounted so that it is free to rotate with negligible friction about an axle that is oriented through the center of the cylinder and perpendicular to the page. A light string is wrapped around the cylinder. At time $t = 0$, the string is pulled with a constant force F_0, which causes the cylinder to rotate, as shown in the figure. After time $t = t_0$, the string is completely unwound from the cylinder and loses contact with it. If necessary, how can a student determine the change in angular momentum ΔL of the cylinder from $t = 0$ to $t = t_0$?

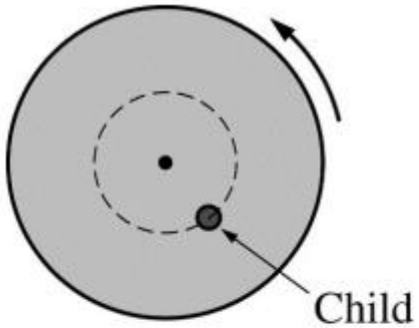
(A) A determination is not necessary. The cylinder does not have a change in angular momentum, because the cylinder does not change its horizontal or vertical position as it rotates.

(B) Multiply F_0, R, and t_0 to calculate the angular impulse. This quantity is equal to ΔL.

(C) Use Newton's second law of motion in the rotational form to determine the angular acceleration of the cylinder. Use the angular acceleration to determine the cylinder's change in angular velocity. Multiply the result by the mass of the cylinder.

(D) Calculate the net torque exerted on the cylinder, then divide the net torque by the cylinder's rotational inertia to find the average angular velocity of cylinder. Multiply the result by the average angular velocity by mass M.

Example Question of AP Classroom



Top View

The diagram above shows a top view of a child of mass M on a circular platform of mass $5M$ that is rotating counterclockwise. Assume the platform rotates without friction. Which of the following describes an action by the child that will result in an increase in the total angular momentum of the child-platform system?

(A) The child moves toward the center of the platform.

(B) The child moves away from the center of the platform.

(C) The child moves along a circle concentric with the platform (dashed line shown) opposite the direction of the platform's rotation.

(D) None of the actions described will change the total angular momentum of the child-platform system.